

$$\begin{array}{ll} \min z = f(x) & x \in \mathbb{R}^m \\ \text{s.t.} & f(x) : \mathbb{R}^m \rightarrow \mathbb{R} \\ & x \in X \end{array}$$

$$\left[\begin{array}{ll} g_i(x) \geq b_i & i = 1 \dots m \\ x_j \geq 0 & j = 1 \dots n \end{array} \right.$$

$f(x)$ linear $g_i(x)$ linear

$$\left[\begin{array}{ll} \min z = c_1 x_1 + \dots + c_n x_n \\ \text{s.t.} & a_{11} x_1 + \dots + a_{1n} x_n \geq b_1 \\ & \vdots \\ & a_{m1} x_1 + \dots + a_{mn} x_n \geq b_m \\ & x_1, \dots, x_n \geq 0 \end{array} \right.$$

$$\max z = \min -z$$

$$\begin{aligned} & a_{11} x_1 + \dots + a_{1n} x_n \leq b_i \\ \Leftrightarrow & -a_{11} x_1 - \dots - a_{1n} x_n \geq -b_i \end{aligned}$$

$$a_1 x_1 + \dots + a_n x_n = b_i$$

$$\begin{aligned} & a_1 x_1 + \dots + a_n x_n \leq b_i \\ & -a_1 x_1 - \dots - a_n x_n \leq -b_i \end{aligned}$$

$$x \geq 0$$

$$x_j \leq 0 \Rightarrow x_j' = -x_j \geq 0$$

$$x_j \in \mathbb{R}$$

$$\Rightarrow x_j = x_j' - x_j''$$

$$3x \quad x \in \mathbb{R}$$

$$3x' - 3x''$$

$$x' \geq 0$$

$$x'' \geq 0$$

$$x' \geq 0$$

$$x'' \geq 0$$

$$c_1 \dots c_m = C^T$$

$$x_1 \dots x_m$$

$$\begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} = b$$

$$[a_{ij}] = A$$

costi

$$\left[\begin{array}{l} \max z = C^T x \\ Ax \leq b \\ x \geq 0 \end{array} \right]$$

$$\left[\begin{array}{l} \min z = C^T x \\ \text{s.t.} \end{array} \right]$$

$$Ax \geq b$$

RISORSE

$$x \geq 0$$

COEFFICIENTI TECNOLOGICI

$$\left[\begin{array}{l} \min z = \sum_{j=1}^m c_j x_j \\ \text{s.t.} \sum_{j=1}^m a_{ij} x_j \geq b_i \quad \forall i=1 \dots m \\ x_j \geq 0 \quad \forall j=1 \dots m \end{array} \right]$$

$$x \in \mathbb{R}_{\geq 0}^m$$

PROGRAMMAZIONE LINEARE

$$x \in \mathbb{Z}_{\geq 0}^n \text{ INTERO}$$

$$x \in B^m = \{0, 1\}$$

$\Rightarrow$

$$x \in X$$



$$x \in X \subseteq \mathbb{Z}^m$$



$$x^* = 1501,2 \rightarrow 1501 \text{ veicoli}$$

$$x_i \in \{0, 1\}$$

$$x_1 + x_2 = 1$$

$$x_1, x_2 \in \{0, 1\}$$

$$\oplus x_{02}$$

$$x_1, x_2 \in [0, 1]$$

$$x_1 = 0,5 \cdot x_1 = 0$$

$$x_2 = 0,5 \cdot x_2 = 1$$

Entrata 100%

$$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} \leq \begin{bmatrix} b \\ \vdots \\ b \end{bmatrix}$$

$$\max x \sum_j c_j x_j$$

$$\begin{matrix} 1 & a_{1j} & \leq b_1 \\ \vdots & \vdots & \vdots \\ m & a_{mj} & \leq b_m \end{matrix}$$

PROBLEMA DI OTTIMIZZAZIONE

1 Raccolta dati

$$A: 30\%$$

$$B: 20\%$$

3 Formulazione del problema
 $\Rightarrow$ modello programmat. matematica.

2 Identificazione del problema

Soluzione del problema
 cosa desiderata
 x_A, x_B

3a Variabili

ESOGENE

PARAMETRI

ENDOGENE

VARIABILI

$$\max 30x_A + 20x_B$$

3b f. obiettivo

$$Ax \geq b$$

3c vincoli del problema

$$x_A + x_B \leq 3$$

$$x_A, x_B \geq 0 \Rightarrow x^* = (0, 0)$$

4 Soluzione del problema

- $z^* = f(x^*) = c^T x^* \leq c^T x \quad : x \in X$
- $\left\{ \begin{array}{l} \text{problema inammissibile} \quad X = \emptyset \\ \text{problema illimitato} \quad \forall x \exists K: c^T x < -K \end{array} \right.$
- $\rightarrow \begin{cases} x_A + x_B \geq 2 \\ x_A + x_B \leq -4 \end{cases}$

5. Validazione della soluzione

$$x_A + x_B = 1,8$$

$$0,5 \leq x_a \leq 1,5$$

$$0,5 \leq x_b \leq 1,5$$

$\Rightarrow$

$$\begin{bmatrix} x_b^* = 1,5 \\ x_a^* = 0,3 \end{bmatrix}$$

1.1.

x_1 Bigcoff x_2 latte x_3 corn flakes

$$\text{Min } 0.5x_1 + 0.4x_2 + 0.7x_3$$

s.v.

$$\left[\begin{array}{l} 452x_1 + 64x_2 + 365x_3 \leq 600 \\ 452x_1 + 64x_2 + 365x_3 \geq 400 \\ 8x_1 + 2.3x_3 \geq 10 \\ 47x_2 + 1.8x_3 \geq 10 \\ x_1, x_2, x_3 \geq 0 \end{array} \right. \quad x^* = \begin{bmatrix} 1.25 \\ 0.21 \\ 0 \end{bmatrix}$$

1.2.

x_1 STD x_2 HiQ

$$p = [12, 16]$$

$$b = \begin{bmatrix} 80 \\ 60 \\ 70 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 4 \\ 4 & 4 \end{bmatrix}$$

$$m=2 \\ n=3$$

$$z^* = C^T(x^*) \geq C^T(x)$$

$$\text{Max } z = 12x_1 + 16x_2$$

s.t.

$$3x_1 + 2x_2 \leq 80$$

$$2x_1 + 4x_2 \leq 60$$

$$4x_1 + 4x_2 \leq 70$$

$$x \geq 0 \quad x^* = (12, 5, 5)^T$$

$$z^* = 260$$

$$\text{Min } C^T x$$

$$z^*$$

LOWER BOUND

UPPER BOUND

GAP
LOWER BOUND

$$x(12, 5)$$

1.3

x_{ij} quantità prodotta attraverso l'arco i, j

$$x(p_i, w_j) \geq 0$$

$$y(w_j, c_k) \geq 0$$

$$p_i \rightarrow w_j \left| \begin{array}{c} x(p_i, w_1) \\ x(p_i, w_2) \\ x(p_2, w_1) \\ \vdots \\ \vdots \end{array} \right| \begin{array}{c} 1 \\ 2 \\ 3 \\ \vdots \\ \vdots \end{array} \quad \min \sum c_{ij} x_{ij}$$

$$\begin{aligned} \text{Min} \quad & x(p_1, w_1) + 6 x(p_1, w_2) + 4 x(p_2, w_1) + \dots \\ & + 3 y(w_1, c_1) + 4 y(w_1, c_2) + \dots + 2 y(w_2, c_3) \end{aligned}$$

s.t.

$$\left[\begin{array}{l} y(w_1, c_1) + y(w_2, c_1) = 50000 \\ y(w_1, c_2) + y(w_2, c_2) = 100000 \\ y(w_1, c_3) + y(w_2, c_3) = 50000 \end{array} \right.$$

$$\left[\begin{array}{l} x(p_1, w_1) + x(p_1, w_2) \leq 150000 \\ x(p_2, w_1) + x(p_2, w_2) \leq 60000 \end{array} \right.$$

$$\left[\begin{array}{l} x(p_1, w_1) + x(p_2, w_1) - y(w_1, c_1) - y(w_1, c_2) - y(w_1, c_3) = 0 \\ x(p_1, w_2) + x(p_2, w_2) - y(w_2, c_1) - y(w_2, c_2) - y(w_2, c_3) = 0 \end{array} \right.$$

P set impianti

W magazzini

C clienti

c'_{ij} $i \in P, j \in W$ costi trasporto

c''_{ij} $i \in W, j \in C$ costi di trasporto

k_i $i \in K$ capacità

d_i $i \in D$ domanda

$$\text{Min } z = \sum_{i \in P} \sum_{j \in W} c'_{ij} x_{ij} + \sum_{i \in W} \sum_{j \in C} c''_{ij} y_{ij}$$

s.t.

$$(1) \quad \sum_{i \in W} y_{ij} = d_j \quad \forall j \in D$$

$$(2) \quad \sum_{j \in W} x_{ij} \leq K_i \quad \forall i \in P$$

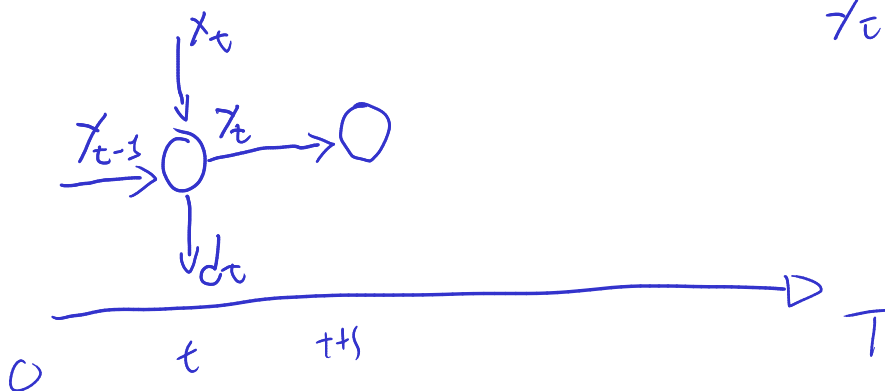
$$(3) \quad \sum_{i \in P} x_{ij} - \sum_{j \in C} y_{ij} = 0 \quad \forall i \in W$$

$$\begin{aligned} x_{ij} &\geq 0 & i \in P, j \in W \\ y_{ij} &\geq 0 & i \in W, j \in C \end{aligned}$$

1.4. $\begin{cases} x_t & \text{quantità prodotta al tempo } t \\ y_t & \text{quantità in scorta al tempo } t \end{cases} \quad T=5$

$$* \quad z = \sum_{t=1}^T c_t x_t + \sum_{t=1}^T h_t y_t$$

~~$x_t \geq 0 \quad t \in \{0, \dots, T\}$~~
 $x_t \geq 0 \quad t \in \{1, \dots, T\}$
 $y_t \geq 0 \quad t \in \{0, \dots, T\}$



s.t.

$$\begin{aligned} (1) \quad & x_t + y_{t-1} - y_t = d_t & \forall t \in 1..T \\ (2) \quad & y_0 = 0 \\ (3) \quad & y_t \leq 30 & \forall t \in 1..T \end{aligned}$$

1.5

$$m x_1 + a = y_1$$

$$m x_2 + a = y_2$$

⋮

100h

+ soln

$$m x_i = y_i - a$$

$$\sum_{j=1}^m a_j x_j \leq b$$

hard constraint

↓
soft constraint + consto

$$\sum_{j=1}^m a_j x_j - \mu \leq b$$

 $\mu \geq 0$ slacks.t. min μ

$$\sum_{j=1}^n a_j x_j + v \geq b \quad v \geq 0$$

$$\begin{cases} \text{Min } \sum_{i=1..m} c' \mu_i + \sum_{i=1..m} c'' v_i \\ \sum_{j=1}^n a_j x_j - \mu_i + v_i = b_i \\ \mu_i \geq 0 \rightarrow c' \\ v_i \geq 0 \rightarrow c'' \end{cases}$$

1.5

$$\text{Min } \sum_{i=1}^{10} \mu_i + \sum_{i=1}^{10} v_i$$

$$\textcircled{*} \begin{cases} m x_i + \mu_i - v_i = y_i - a \quad \forall i \in 1..10 \\ \mu_i \geq 0 \\ v_i \geq 0 \end{cases}$$

$$\begin{aligned} 2.7m + \mu_1 - V_1 &= 12 - Q \\ 0.7m + \mu_2 - V_2 &= 6 - Q \\ &\vdots \\ 2m + \mu_{10} - V_{10} &= 12.2 - Q \end{aligned} \quad \begin{aligned} &(m^*, Q^*, u_i^*, v_i^*) \\ &\vec{z}, \vec{x} \end{aligned}$$

1.5. b

f.o. $z = \min \max_{i \in 1..10} \{u_i, v_i\}$

introduco una variabile scalare z

Min z
s.t. $\begin{bmatrix} \end{bmatrix}$



$$\begin{array}{ll} m_i \leq z & \forall i \in 1 \dots 10 \\ v_i \leq z & \forall i \in 1 \dots 10 \end{array}$$

1.5.c. fitting quadratic

$$y = cx^2 + mx + a$$

$$c(x_i)^2 + mx_i + a = y_i$$

$$C(X_i)^2 + mX_i + M_i - V_i = Y_i - Q \quad \forall i = 1, \dots, 10$$

$$C(2,7)^2 + m \cdot 2,7 + u_i - v_i = 12 - \alpha$$

⋮

$$1) \left[\text{Min} \sum u_i + \sum v_i \right]$$

$$2) \left[\begin{array}{l} \text{Min } z \\ u_i \leq z \\ v_i \leq z \end{array} \right]$$

$$\left[\begin{array}{l} \text{Min} \sum_{j=1}^m c_j x_j \\ \sum_{j=1}^m a_{ij} x_j \geq b_i \quad \forall i = 1 \dots m \\ x_j \geq 0 \end{array} \right]$$

Costo di ATTIVAZIONE (SETUP) F_j

$$\left[z_j = \begin{array}{ll} c_j x_j + F_j & \text{se } x_j > 0 \\ 0 & \text{se } x_j = 0 \end{array} \right]$$

$$\left[\begin{array}{l} y_j \in \{0, 1\} \quad \begin{array}{ll} y_j = 0 & x_j = 0 \\ y_j = 1 & x_j > 0 \end{array} \\ \text{Min} \sum_j c_j x_j + \sum_j F_j y_j \\ \sum_j a_{ij} x_j \geq b_i \\ x_j \leq M_j y_j, \quad \forall j \in 1 \dots m \\ x_j \geq 0, \quad y_j \in \{0, 1\} \end{array} \right] \quad \left[\begin{array}{l} \text{M sufficiently} \\ \text{grande} \end{array} \right]$$

(z^m)